

UChicago REU Notes Week 2

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1 Review: Symmetric Monoidal Categories

Definition 1.1. A category $\mathcal{V}$ is a *monoidal category*, if there is a bifunctor $\otimes : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$ (called *monoidal product*), and an object $\kappa \in \mathcal{V}$ (called *unit/monoidal unit*), together with the following natural isomorphisms:

- The *associator* $a : (- \otimes -) \otimes - \xrightarrow{\sim} - \otimes (- \otimes -)$ of functors $\mathcal{V} \times \mathcal{V} \times \mathcal{V} \xrightarrow{\sim} \mathcal{V}$. (So, $(X \otimes Y) \otimes Z \cong X \otimes (Y \otimes Z)$ in a canonical way).
- The *left unit* $\lambda : \kappa \otimes - \xrightarrow{\sim} \text{Id}_{\mathcal{V}}$ ($1 \otimes X \cong X$ in a canonical way).
- The *right unit* $\rho : - \otimes \kappa \xrightarrow{\sim} \text{Id}_{\mathcal{V}}$ ($X \otimes 1 \cong X$ in a canonical way).

They also satisfy two extra axioms:

- *Triangle Axiom:*

$$\begin{array}{ccc} (X \otimes 1) \otimes Y & \xrightarrow{a_{X,1,Y}} & X \otimes (1 \otimes Y) \\ & \searrow \rho_X \otimes \text{id}_Y \quad \swarrow \text{id}_X \otimes \lambda_Y & \\ & X \otimes Y & \end{array}$$

- *Pentagon Axiom:*

$$\begin{array}{ccc} ((X \otimes Y) \otimes Z) \otimes W & \xrightarrow{a_{X \otimes Y, Z, W}} & (X \otimes Y) \otimes (Z \otimes W) \\ \downarrow a_{X,Y,Z} \otimes \text{id}_W & & \downarrow a_{X,Y,Z \otimes W} \\ (X \otimes (Y \otimes Z)) \otimes W & & X \otimes (Y \otimes (Z \otimes W)) \\ \searrow a_{X,Y \otimes Z, W} & \nearrow \text{id}_X \otimes a_{Y,Z,W} & \\ & X \otimes ((Y \otimes Z) \otimes W) & \end{array}$$

Based on *MacLane's Coherence Theorem*, the above two axioms guarantees all finite sequences of a, λ, ρ and their inverses agree, namely there is a unique isomorphism $P_1 \rightarrow P_2$, where P_i are (possibly) different

orders of performing product on $X_1, \dots, X_n \in \mathcal{V}$. For reference, cf. *Category Theory for Working Mathematicians*.

This guarantees all monoidal category to be equivalent to a *strict monoidal category*, namely all a, λ, ρ above are “identity transformations”, or $1 \otimes X = X, X \otimes 1 = X$, and $(X \otimes Y) \otimes Z = X \otimes (Y \otimes Z)$, so WLOG one can assume the product $\otimes$ is associative.

Definition 1.2. A monoidal category $\mathcal{V}$ is *braided*, if there is a natural isomorphism $B : {}_{-1} \otimes {}_{-2} \xrightarrow{\sim} {}_{-2} \otimes {}_{-1}$ (so $B_{X,Y} : X \otimes Y \xrightarrow{\sim} Y \otimes X$ is an isomorphism), together with the *Hexagon axioms* satisfied:

$$\begin{array}{ccc}
(X \otimes Y) \otimes Z & \xrightarrow{a_{X,Y,Z}} & X \otimes (Y \otimes Z) \xrightarrow{B_{X,Y \otimes Z}} (Y \otimes Z) \otimes X \\
\downarrow B_{X,Y} \otimes \text{Id}_Z & & \downarrow a_{Y,Z,X} \\
(Y \otimes X) \otimes Z & \xrightarrow{a_{Y,X,Z}} & Y \otimes (X \otimes Z) \xrightarrow{\text{Id}_Y \otimes B_{X,Z}} Y \otimes (Z \otimes X) \\
\\
X \otimes (Y \otimes Z) & \xleftarrow{a_{X,Y,Z}} & (X \otimes Y) \otimes Z \xrightarrow{B_{X \otimes Y,Z}} Z \otimes (X \otimes Y) \\
\downarrow \text{Id}_X \otimes B_{Y,Z} & & \uparrow a_{Z,X,Y} \\
X \otimes (Z \otimes Y) & \xleftarrow{a_{X,Z,Y}} & (X \otimes Z) \otimes Y \xrightarrow{B_{X,Z} \otimes \text{Id}_Y} (Z \otimes X) \otimes Y
\end{array}$$

$\mathcal{V}$ is a *symmetric monoidal category*, if $B_{Y,X} = B_{X,Y}^{-1}$ for all $X, Y \in \mathcal{V}$.

The two hexagons are guaranteeing the ability to perform 3-cycle permutations on three consecutive objects in a product, hence together with B serving as transpositions, it guarantees isomorphism between all permutations of finite products. The symmetric axiom further imposes that there is a single way of doing such permutation (so there’s no “nontrivial braiding”, when $B_{Y,X} \circ B_{X,Y} : X \otimes Y \rightarrow Y \otimes X \rightarrow X \otimes Y$ is not identity). From my naive understanding of coherence theorem, WLOG we can again treat $X \otimes Y = Y \otimes X$.

Example 1.3. Everyone loves unital commutative ring R , so I use category of R -modules as example :)

Equipped with tensor product $\otimes_R$, and R itself as tensor unit (up to isomorphism), we see $R\text{-Mod}$ forms a symmetric monoidal category due to following natural isomorphisms:

$$\left\{ \begin{array}{ll} (M \otimes_R N) \otimes_R L \cong M \otimes_R (N \otimes_R L) & , \quad (m \otimes n) \otimes \ell \mapsto m \otimes (n \otimes \ell) \\ R \otimes_R M \cong M & , \quad r \otimes m \mapsto rm \\ M \otimes_R R \cong M & , \quad m \otimes r \mapsto rm \\ M \otimes_R N \cong N \otimes_R M & , \quad m \otimes n \mapsto n \otimes m \end{array} \right.$$

Now, observe that $R\text{-Mod}$ is self-enriched, meaning $\text{Hom}_R(-, -) : R\text{-Mod}^{\text{op}} \times R\text{-Mod} \rightarrow R\text{-Mod}$ is a bifunctor (also called an *internal hom functor*). Moreover, there’s the tensor-hom adjunction:

$$\text{Hom}_R(M \otimes_R N, L) \cong \text{Hom}_R(M, \text{Hom}_R(N, L)), \quad f \mapsto (m \mapsto f(m \otimes -))$$

This indicates $(- \otimes_R N \dashv \text{Hom}_R(N, -))$ is an adjoint pair, guaranteeing $- \otimes_R N$ commutes with all colimits. For instance, $(M \oplus L) \otimes_R N = (M \otimes_R N) \oplus (L \otimes_R N)$. This gives rise to the following definition:

Definition 1.4. $\mathcal{V}$ is a *symmetric monoidally closed category*, if there exists an *internal hom functor* $\mathcal{V}(-, -) : \mathcal{V}^{\text{op}} \times \mathcal{V} \rightarrow \mathcal{V}$, such that the adjunction $(- \otimes X \dashv \mathcal{V}(X, -))$ holds for all $X \in \mathcal{V}$.

$\mathcal{V}$ is a *symmetric monoidally cocomplete category*, if it’s cocomplete, together with $- \otimes X, X \otimes -$ preserves colimits for all $X \in \mathcal{V}$.

If people knows about Yoneda Lemma, welcome to look into the last section, which stregnthen the isomorphism in monoidal closeness to natural isomorphism $\mathcal{V}(- \otimes X, -) \cong \mathcal{V}(-, \mathcal{V}(X, -))$ on the bifunctors $\mathcal{V}^{\text{op}} \times \mathcal{V} \rightarrow \mathcal{V}$.

Example 1.5. The category of unital commutative rings $\mathbf{CRing}$ with $\otimes_{\mathbb{Z}}$ as monoidal product is in fact not monoidally closed. Suppose the contrary there exists such bifunctor $C : \mathbf{CRing}^{\text{op}} \times \mathbf{CRing} \rightarrow \mathbf{CRing}$, such that the adjunction holds, then the following isomorphism holds for any rings $R, S \in \mathbf{CRing}$:

$$\text{Hom}_{\mathbf{CRing}}(R, S) \cong \text{Hom}_{\mathbf{CRing}}(\mathbb{Z} \otimes_{\mathbb{Z}} R, S) \cong \text{Hom}_{\mathbf{CRing}}(\mathbb{Z}, C(R, S)) = \{*\}$$

The final implication follows from the fact $\mathbb{Z} \in \mathbf{CRing}$ is initial. Yet, here's a counterexample: $\text{Hom}_{\mathbf{CRing}}(\mathbb{C}, \mathbb{C})$ as at least two maps, including identity ($z \mapsto z$) and conjugation ($z \mapsto \bar{z}$), so we reach a contradiction.

Notice this is distinct from $\mathbf{CRing}$ being monoidally cocomplete: Coequalizer exists in $\mathbf{CRing}$ (by quotienting out the ideal generated by difference of outputs), and $\otimes_{\mathbb{Z}}$ is the coproduct in $\mathbf{CRing}$, hence all colimits exist (cf. *Limit Existence Theorem*). Moreover, colimits commutes with all colimit, in particular so does $\otimes_{\mathbb{Z}}$.

Remark 1.6. From now on, we assume either $\mathcal{V}$ is monoidally cocomplete. If suitable, also assume internal hom functor exists.

One reason to assume this, is to make sure some of the construction makes sense: In the literature professor May (implicitly) assumes monoidal product commutes with colimits, and occasionally use internal hom as an example of operad.

2 Intro: Operads

Notation 2.1. Let Σ_j denotes the j th symmetric group for all $j \in \mathbb{N}$. In particular, $\Sigma_0 = \Sigma_1 = \{1\}$ as trivial group.

Let Σ denote the category of finite numbers, with objects being $\mathbf{j} := \{1, \dots, j\}$, and homset

$$\mathrm{Hom}_{\Sigma}(\mathbf{i}, \mathbf{j}) = \begin{cases} \emptyset & , \quad i \neq j \\ \Sigma_j & , \quad i = j \end{cases}$$